Lecture no. 5:

Central Tendency, Median, Quartiles, Box Plots

Problem:

Average :-

mean

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
3	4	9	17	18	20	22	23	29	32	39	42	42	42	51	47

min　　max

Arithmetic mean = 30.6

if 47 become 197
the mean will increase
to 36.01 where is 47 become
so mean will become 23.0
Therefore 47 is $\boxed{\text{outlier}}$

Do sorting of Average array :-

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
3	4	9	17	18	20	22	23	29	32	39	42	42	42	51	47

min　　max

$\underbrace{3\ 4\ 9\ 17\ 18\ 20\ 22\ 23}_{8}$ median $\underbrace{29\ 32\ 39\ 42\ 42\ 42\ 51\ 47}_{8}$

$$\text{median} = \frac{23+29}{2} = 26$$

Median are "Robust to outliers"

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
3	4	9	17	18	20	22	23	29	32	39	42	42	42	51	47

$\overbrace{\overbrace{3\ 4\ 9\ 17}^{4}\ \overbrace{18\ 20\ 22\ 23}^{4}}^{8}$ $\overbrace{\overbrace{29\ 32\ 39\ 42}^{4}\ \overbrace{42\ 42\ 51\ 47}^{4}}^{8}$

min　　max

Q1
0.25

Median
Q2
0.5

Q3
0.75

→ Quartiles
Quantiles

Computing quantiles:-

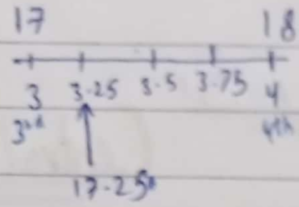
$$0.25 \rightarrow \left[0.25\,(\underbrace{16}_{\text{total no. of values}}+1) \right]^{th} \text{ value}$$

4.25^{th} Value

-1 (0-based index)

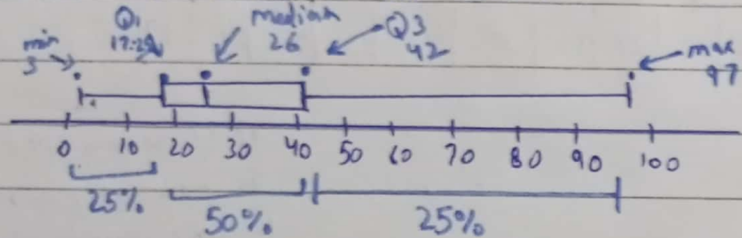
3.25^{th} value

$\boxed{\text{quantile}(0.25) = 3.25^{th} \text{ value} = 17.25}$



Magnitude of values is not clear!

So,



We have done 3 type of averages :-

1. Arithematic Mean (Average)
2. Median (Mid value)
3. Mode (Most frequent value in set).

In this lecture all code is done under file name is "02-studying-variables.ipynb".